

Software Outline Document

Overview

This system is made up of two Arduino software components:

Sound Logging System: Reads microphone input, converts the signal to decibel values, and logs the data to an SD card when a trigger is active.

LED Flicker System: Monitors a trigger input and microphone level, then generates randomized flickering LED patterns to create a disorienting strobe effect.

Software Files

File: sound_logger.ino

Purpose: Logs sound data (time, raw value, and dB) to an SD card when the trigger pin reads HIGH.

File: led_flicker.ino

Purpose: Controls LED flickering patterns on pin 2 based on trigger and microphone input. Pin setup covers pins 2–13 as outputs, but all active flicker patterns use pin 2.

Functions

Sound Logger Functions

setup()

- Serial communication at 9600 baud
- Trigger pin (A0) and microphone pin (A2) as inputs
- SD card on chip select pin 10
- Removes the old log file if it exists (log.txt)
- Creates a new log file with a CSV header: Time(ms), Raw, dB

loop()

- Checks if the trigger pin (A0) is HIGH
- If HIGH, then reads the microphone value, calculates dB, opens the log file, writes a data row, and closes the file

- Adds a 200ms delay between readings

LED Flicker Functions

setup()

- Sets pins 2 through 13 as outputs (for LEDs)
- Sets A0 as input (trigger pin)
- Sets A1 as input (microphone pin)

loop()

- Waits until trigger pin (A0) reads HIGH
- While microphone signal (A1) reads below 500:
 - Randomly selects a number from 0–9 using random(0,9)
 - Executes the matching flicker pattern with switch statement
 - Each case calls flicker() three times with different durations, separated by delay() calls

flicker(int thisPin, int duration, int brightness)

- Creates a strobe effect on the specified LED pin
- Parameters:
 - thisPin- the LED pin number to flash
 - duration- total flicker time in milliseconds
 - brightness- LED brightness level (0–255)
- Operation: loops duration / 70 times, turning the LED on then off with 35ms delays each cycle

Key Variables

Sound Logger Variables

Variable	Type	Description
micPin	Const int	Microphone input pin, set to A2
triggerPin	Const int	Trigger input pin, set to A0
chipSelect	Const int	SD card chip select pin, set to 10
myFile	file	File object used to open and write to log.txt
sensorValue	int	Raw analog reading from microphone (0–1023)
dB	float	Calculated approximate sound level in decibels

LED Flicker Variables

Variable	Type	Description
lowestPin	Const int	First LED pin in range, set to 2
highestPin	Const int	Last LED pin in range, set to 13
thisPin	int	Declared globally. Used as a local parameter inside flicker()
duration	int	Declared globally. Used as a local parameter inside flicker()

brightness	int	Declared globally. Used as a local parameter inside flicker()
freq	float	Set to 35 (timing reference)

Data Processing

Sound Logger:

Converts the analog microphone reading into an approximate dB value using:

$$\text{dB} = 30 + (\text{sensorValue}/1023.0) \times 104$$

Each log entry stores time in milliseconds, raw sensor value, and calculated dB.

LED Flicker System:

Uses random(0, 9) to select one of 10 switch-case patterns. Each pattern calls flicker() three times with varying durations (200–800ms) and fixed delays between calls to produce irregular strobing.

System Behavior Summary

Sound Logger

- Initialize SD card and create log file
- Wait for trigger pin to go HIGH
- Read microphone, calculate dB, write to file
- Wait 200ms and repeat

LED Flicker System

- Wait for trigger pin (A0) to go HIGH
- While microphone (A1) reads below threshold (500):
 - Pick random flicker pattern
 - Execute three flicker sequences with delays
- Stop flickering when microphone exceeds threshold or trigger goes LOW

Possible Improvements

- Calibrate microphone for accurate dB readings against a reference
- Replace delay() calls with non-blocking timing using millis()
- Add input filtering to the microphone reading (average)
- Buffer SD writes instead of opening/closing the file in every loop iteration